

9.8.32

AI25BTECH11030 -Sarvesh Tamgade

Question: The centre of the circle passing through the point $(0, 1)$ and touching the curve $y = x^2$ at $(2, 4)$ is

- (a) $\left(\frac{-16}{5}, \frac{27}{10}\right)$
- (b) $\left(\frac{-16}{7}, \frac{53}{10}\right)$
- (c) $\left(\frac{-16}{5}, \frac{53}{10}\right)$
- (d) none of these

Solution:

Input Variables:

Parameters of the Circle:

Where $\mathbf{c}$ is the centre of circle.

$$\mathbf{V}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_1 = -\mathbf{c} \quad (4.1)$$

Parameters of the Parabola:

$$\mathbf{V}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, f_2 = 0 \quad (4.2)$$

$$\mathbf{P} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{Q} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad (4.3)$$

Step 1: Find one equation for $\mathbf{c}$

Points P and Q lie on the circle so,

$$\mathbf{P}^\top \mathbf{V}_1 \mathbf{P} - 2\mathbf{c}^\top \mathbf{P} + f_1 = 0, \quad (4.4)$$

$$\mathbf{Q}^\top \mathbf{V}_1 \mathbf{Q} - 2\mathbf{c}^\top \mathbf{Q} + f_1 = 0, \quad (4.5)$$

Subtracting (0.4)-(0.5),

$$(\mathbf{P}^\top \mathbf{V}_1 \mathbf{P} - \mathbf{Q}^\top \mathbf{V}_1 \mathbf{Q}) - 2\mathbf{c}^\top (\mathbf{P} - \mathbf{Q}) = 0, \quad (4.6)$$

which gives the relation:

$$2\mathbf{c}^\top (\mathbf{P} - \mathbf{Q}) = \mathbf{P}^\top \mathbf{V}_1 \mathbf{P} - \mathbf{Q}^\top \mathbf{V}_1 \mathbf{Q}. \quad (4.7)$$

$$-2\mathbf{c}^\top \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix}. \quad (4.8)$$

$$\begin{pmatrix} 4 & 6 \end{pmatrix} \mathbf{c} = 19 \quad (4.9)$$

Step 2: Find the equation of tangent through (2,4) at Parabola

$$(\mathbf{V}_2 \mathbf{Q} + \mathbf{u}_2)^\top \mathbf{x} + \mathbf{u}_2^\top \mathbf{Q} + f_2 = 0, \quad (4.10)$$

$$\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} \right)^\top \mathbf{x} + \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}^\top \begin{pmatrix} 2 \\ 4 \end{pmatrix} + 0 = 0, \quad (4.11)$$

$$\left(2 \quad -\frac{1}{2} \right) \mathbf{x} = 2. \quad (4.12)$$

Step 3: Find the equation of normal to circle at (2,4)

Let the equation of normal to circle at (2,4) be:

$$\mathbf{n}^\top \mathbf{x} = 1 \quad (4.13)$$

The normal is perpendicular to the previously calculated tangent in (0.12)

$$\mathbf{n}^\top \begin{pmatrix} 2 \\ -\frac{1}{2} \end{pmatrix} = 0 \quad (4.14)$$

It passes through (2,4)

$$\mathbf{n}^\top \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 1 \quad (4.15)$$

Row Reduction Steps:

$$\left(\begin{array}{cc|c} 2 & -\frac{1}{2} & 0 \\ 2 & 4 & 1 \end{array} \right) \quad (4.16)$$

Perform the row operation $R_2 \rightarrow R_2 - R_1$:

$$\left(\begin{array}{cc|c} 2 & -\frac{1}{2} & 0 \\ 0 & \frac{9}{2} & 1 \end{array} \right) \quad (4.17)$$

Perform the row operation $R_1 \rightarrow R_1 + \frac{R_2}{9}$:

$$\left(\begin{array}{cc|c} 2 & 0 & \frac{1}{9} \\ 0 & \frac{9}{2} & 1 \end{array} \right) \quad (4.18)$$

$$\mathbf{n} = \frac{1}{18} \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad (4.19)$$

Equation of normal is :

$$(1 \ 4) \mathbf{x} = 18 \quad (4.20)$$

Step 4: Find $\mathbf{c}$

Centre lies on the normal (4.20)

$$(1 \ 4) \mathbf{c} = 18 \quad (4.21)$$

from equation (4.9)

$$(4 \ 6) \mathbf{c} = 19 \quad (4.22)$$

Row reduction steps:

$$\left(\begin{array}{cc|c} 1 & 4 & 18 \\ 4 & 6 & 19 \end{array} \right) \quad (4.23)$$

Perform the row operation $R_2 \rightarrow R_2 - 4R_1$:

$$\left(\begin{array}{cc|c} 1 & 4 & 18 \\ 0 & -10 & -53 \end{array} \right) \quad (4.24)$$

Perform the row operation $R_1 \rightarrow R_1 + \frac{2}{5}R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & -\frac{16}{5} \\ 0 & -10 & -53 \end{array} \right) \quad (4.25)$$

$$\mathbf{c} = \left(-\frac{16}{5}, \frac{53}{10} \right) \quad (4.26)$$

Final Answer:

The centre of the circle is $\left(-\frac{16}{5}, \frac{53}{10} \right)$.

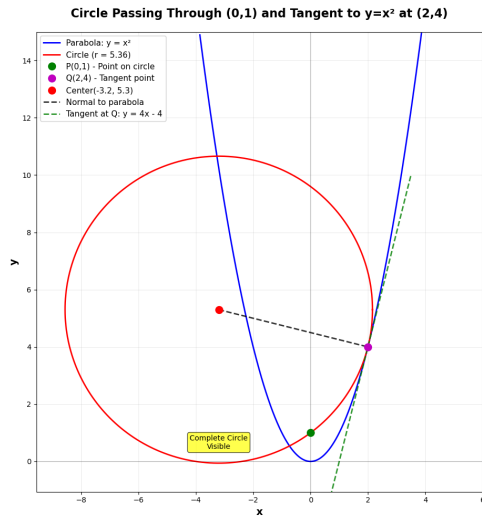


Fig. 4.1: Vector Representation